Carolin Walter, Sarah Sandmann and Julian Varghese
Institute of Medical Informatics, University of Münster, Germany

Institut für Medizinische Informatik

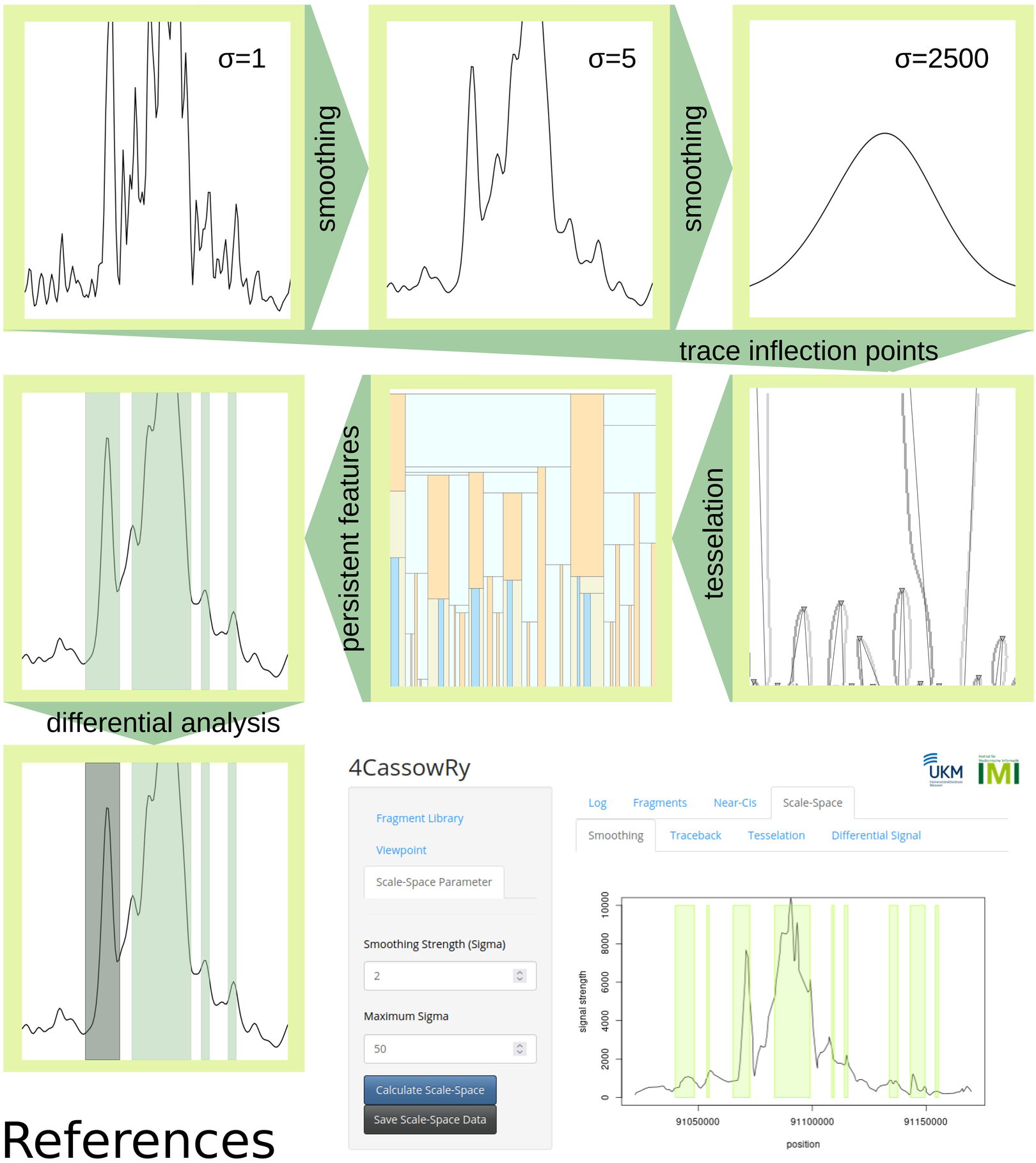
4CassowRy: 4C-seq-applied scale-space transformation with R/Shiny

Introduction

Circular Chromosome conformation capture with sequencing (4C-seq) is a specialized next-generation sequencing technique that allows to explore spatial contacts for genomic loci of interest with high resolution [1]. Due to 4C-seq's fragmented and semi-quantitative data structure, filtering steps as well as smoothing and window approaches are frequently employed to decrease noise levels and assist in the interpretation of the actual signal [2]. Depending on the strength and structure of the actual interactions, arbitrary parameter choices may hinder the overall analysis, and multi-scale solutions can therefore offer a more comprehensive understanding of the interaction landscape at a chosen locus.

R/Shiny App

The R/Shiny app 4CassowRy, “4C-seq-applied scale-space transformation with R/Shiny” includes functionality for quality control, near-cis visualizations [3] and a multi-scale representation for 4C-seq data based on Witkin's scale-space transformation [4]. Multiple Gaussian smoothing operators with increasing strengths σ create signal curves and traceable inflection points, and thus allow to differentiate between more stable 4C-seq features (“peaks”), and background noise. Multi-scale feature representation and extraction of persistent features allows to explore regions of interest and test for differential signal between conditions with DESeq2 [5], if at least two replicate samples per condition are available. Export functions for scale-space interval data and differential analyses are included.



Discussion and Conclusion

Utilizing R/Shiny's flexibility, 4CassowRy offers interactive, multi-scale analysis options for 4C-seq near-cis regions. Stable features with a high degree of persistence during scale-space transformation can be explored and subsequently tested for significant differences between two conditions. Hence, 4CassowRy can help to facilitate 4C-seq near-cis analyses for both single-sample-based settings and more complex setups.

References

[1] van de Werken et al. (2012) Robust 4C-seq data analysis to screen for regulatory DNA interactions, Nature Methods, 9, 969-972. [2] van de Werken et al. (2012) 4C technology: protocols and data analysis, Methods Enzymology, 513, 89-112. [3] Walter et al. (2014) Basic4Cseq: an R/Bioconductor package for analyzing 4C-seq data, Bioinformatics, 30, 3268–3269. [4] Witkin (1983) Scale-space Filtering, Proceedings of the Eighth International Joint Conference on Artificial Intelligence - Volume 2, IJCAI’83, 1019-1022. [5] Love et al. (2014) Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2, Genome Biology, 15, 550.

UKM Universitätsklinikum Münster

Universität Münster

medizinische fakultät Westfälische Wilhelms-Universität Münster